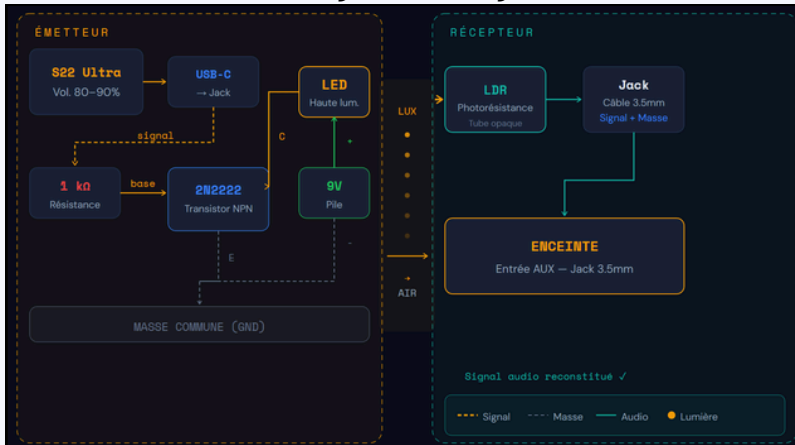
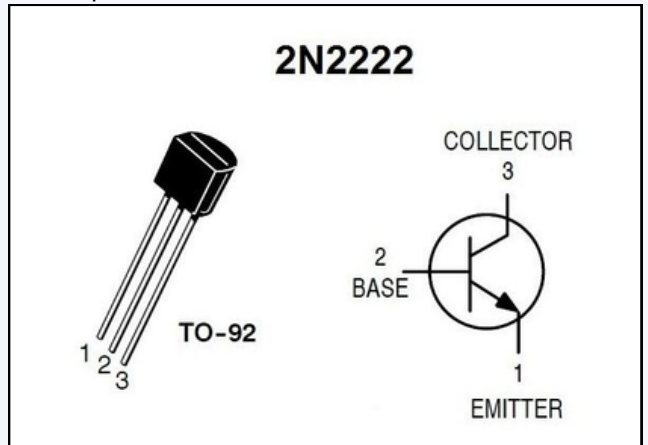


TECHNICAL DATA SHEET (FOR EXPERTS / EDUCATORS) – WORKSHOP 5 : OPTOELECTRONICS & SIGNAL MODULATION

Generic Diagram of a Light Phone



Excerpt from the 2N2222 transistor datasheet



Background : This educational workshop focuses on the high-speed wireless transmission of analog information via directional optical pathways (Li-Fi technology / free-space telecommunications).

Objectives of the experiment : to modulate the light intensity of a source using an audio signal from a smartphone, transmit this signal through the air, and demodulate it using a semiconductor sensor connected to a speaker.

I. Physical Phenomena in Focus

- Analog current modulation (transmitter) : The alternating analog audio signal from the smartphone's headphone jack is fed into the base of a switching transistor (2N2222). This transistor modulates the direct current I_f flowing through the LED or laser diode. The emitted optical power $P_{opt}(t)$ thus varies linearly in exact step with the voltage variations $V_{audio}(t)$ of the audio signal :

$$P_{opt}(t) \propto I_f(t) = \beta \cdot I_b(t)$$

- Photoresistive effect (receiver) : The receiving sensor is a photoresistor (LDR—Light Dependent Resistor). When incident photons strike the LDR's semiconductor material, they release charge carriers (electron-hole pairs) via the internal photoelectric effect. The component's electrical resistance drops drastically as a function of illuminance E (semiconductor law) : $R_{LDR} \propto E^{-\alpha}$
- Demodulation and acoustic transduction : Changes in light intensity detected by the LDR instantly alter the impedance of the audio speaker's input circuit. This demodulated alternating electrical signal is then amplified and converted into mechanical sound pressure waves by the speaker diaphragm, reproducing the original music.

II. Experimental Protocol and Technical Alignment

- Axial optical alignment : Position the emission axis of the high-brightness LED or laser in perfect line-of-sight alignment with the active surface of the photoresistor to maximize the signal-to-noise ratio (SNR).
- Stray Light Isolation : Set up the assembly in a low ambient light environment or use an opaque tube around the LDR to prevent light from ceiling fluorescent lights from saturating the semiconductor sensor.

DID YOU KNOW THAT ?

Did you know ? You can send your music traveling along an invisible thread of light ! Did you know that music can turn into light and travel through the air ? In this workshop, when you pick up your phone, it sends musical notes in the form of electrical pulses to an LED. This light flashes so quickly (thousands of times per second) that your eyes perceive it as simply being on ! Across from it, an electronic sensor called a photoresistor "reads" these light vibrations and instantly converts them back into music in the speaker. If you wave your hand through the middle, you block the beam and the music stops instantly ! This is the principle behind **Li-Fi** and **fiber-optic** networks, which today carry the Internet at the speed of light across the entire planet (Academic source: *IEEE Photonics Technology Letters*).